

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

WORK, POWER, ENERGY

LECTURE-4

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1 Revision
- 2 Inelastic Collision ✓
- 3 Power ✓
- 4 Gabbar Numerical Series
- 5 ..

**SUCCESS IS NOT PERMANENT AND
FAILURE IS NOT FINAL
NEVER STOP WORKING AFTER SUCCESS
AND, NEVER STOP TRYING AFTER
FAILURE**

Recap From 1st Lecture

Work

- mode of transfer of energy

$$W = \vec{F} \cdot \vec{x}$$

$$W = Fx \cos \theta$$

Work

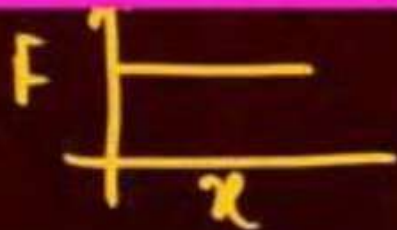
W = +ive

W = -ive

W = 0

Work

Work done by constant



$$W = \vec{F} \cdot \vec{x}$$

Work done by variable force



$$W = \int F \cdot dx$$

Energy

- Capacity to do work

$$m \cdot E$$

K.E

+

P.E

$$K.E = \frac{1}{2}mv^2$$

(Derivation)

✓
✓
✓
Detail
next
Lecture

Recap From 2nd Lecture

• Work-Energy Theorem

$$W = \Delta K \cdot E$$

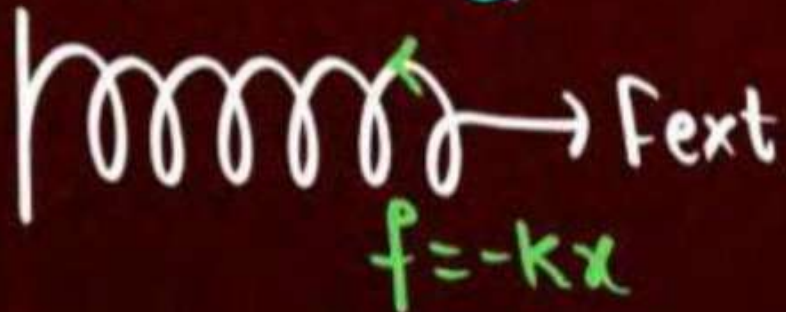
• P.E / U → Potential Energy

Zabardasti ✓

Relative position of particle

✓ Gravitation $P.E = mgh$

✓ Spring $P.E = \frac{1}{2} kx^2$



• WORK done by spring force

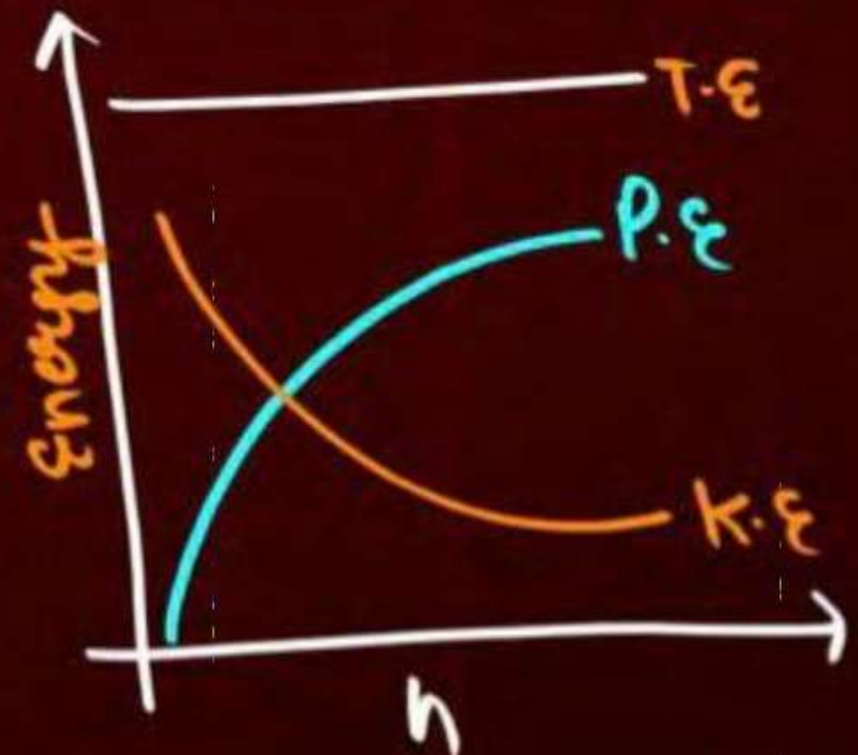
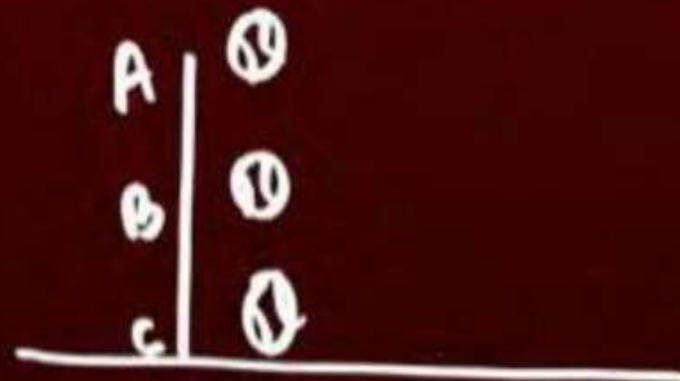
$$W = -\frac{1}{2} kx^2$$

Law of Conservation of Energy

$$T.E_i = T.E_f$$

$$m.E_i = m.E_f$$

100J 100J



Recap From 3rd Lecture

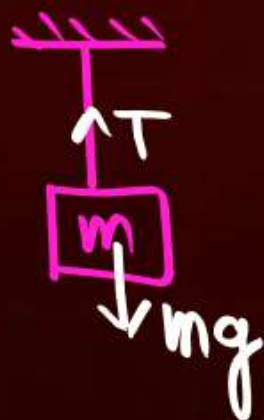
• Conservative Force

- Work $\downarrow$ ^{do not} depend on path.
- ✓ Gravitation Force ✓
- ✓ Spring / Electrostatic Force

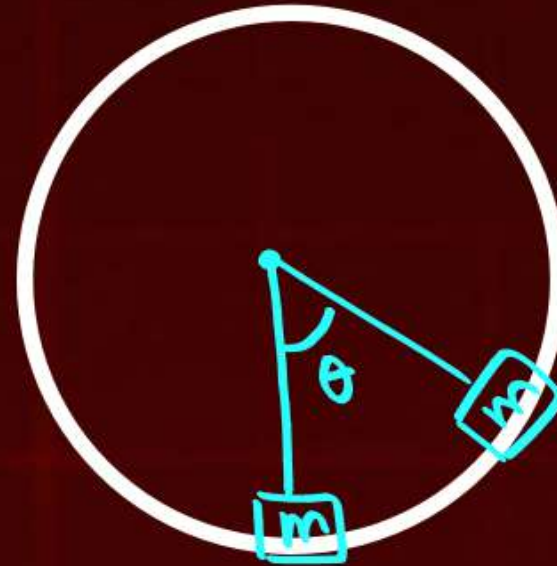
• Non-conservative force

- Work does depend on path
- ✓ Friction

Tension ✓



• Vertical Circular Motion



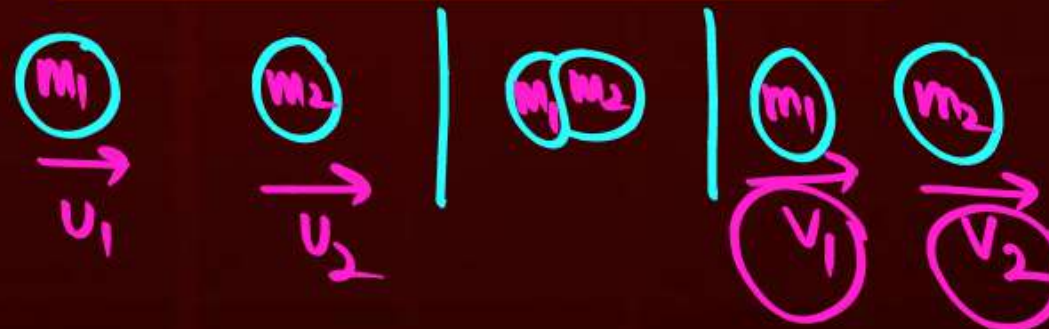
$$F_c = \frac{mv^2}{R} = T - mg \cos \theta$$

$$T = \frac{mv^2}{R} + mg \cos \theta$$

Collision

$P \rightarrow \text{cons.}$ $K.E \rightarrow \text{conserve}$

Perfectly elastic Collision



Perfectly inelastic collision

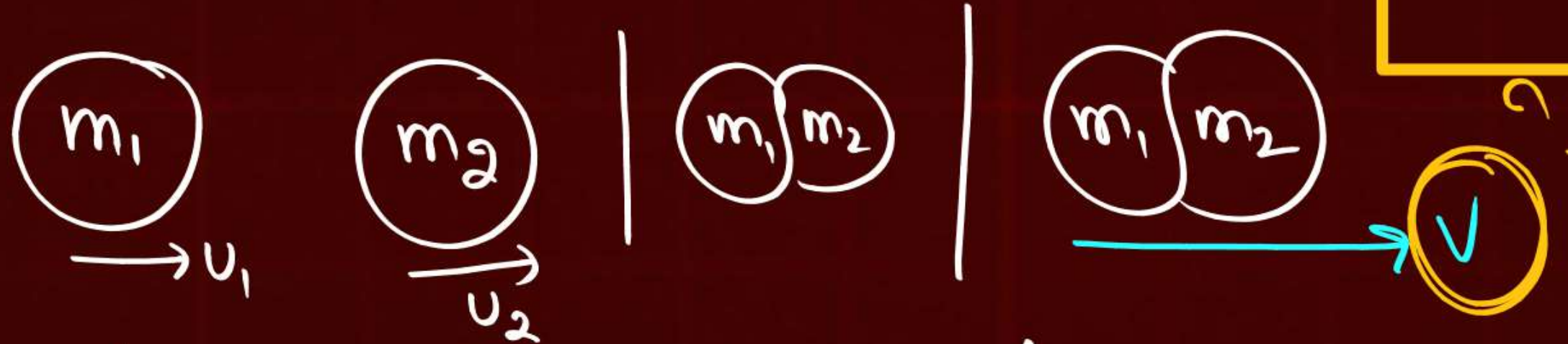
1 on 1

Inelastic Collision

perfectly inelastic collision

During this collision

$p = mv$ only momentum is conserved



only momentum is conserved

$$P_i = P_f$$

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) V$$

$$\frac{m_1 u_1 + m_2 u_2}{m_1 + m_2} = V$$

$$V = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

ch-13

Coefficient of Restitution :- (e)

It is Ratio of (Velocity of Separation) to the velocity of approach ($v_2 - v_1$)

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$v - v - v - v - v - v - v$
Imp

$e = 0$ (Perfectly inelastic collision)

$e = 1$ (Perfectly elastic collision)

Numerical

Q 1

m_1

$$u_1 = 10 \text{ m/s}$$

m_2

$$u_2 = 5 \text{ m/s}$$



$$V_1 = 10$$

$$V_2 = 10$$

m_1 m_2



$$V = 10 \text{ m/s}$$

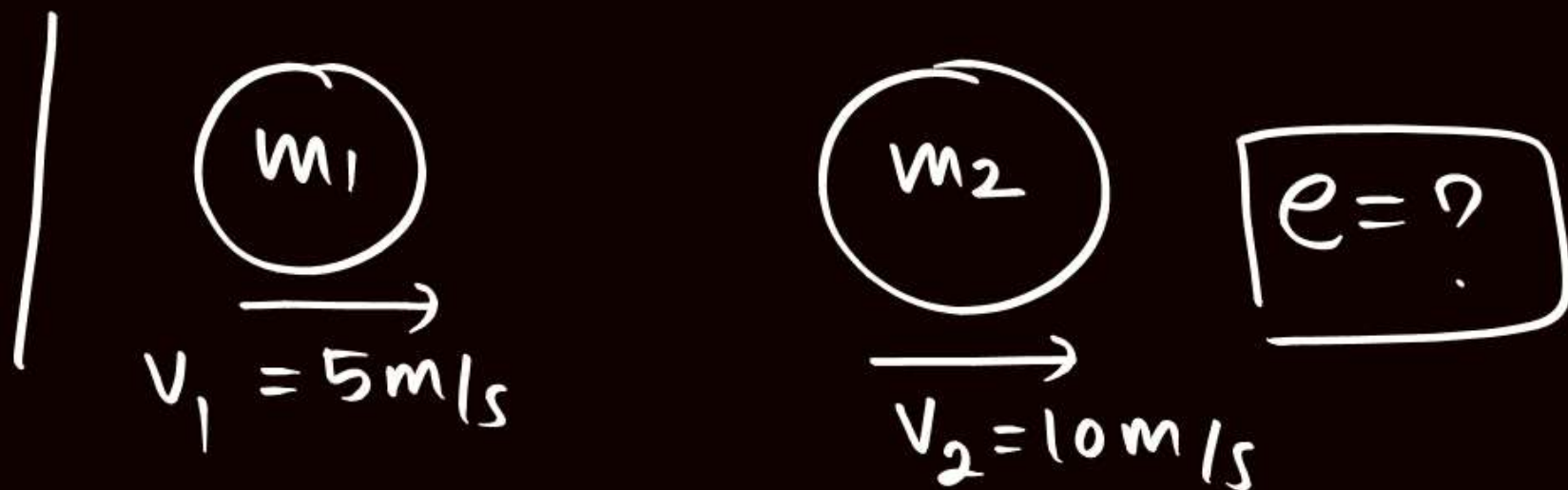
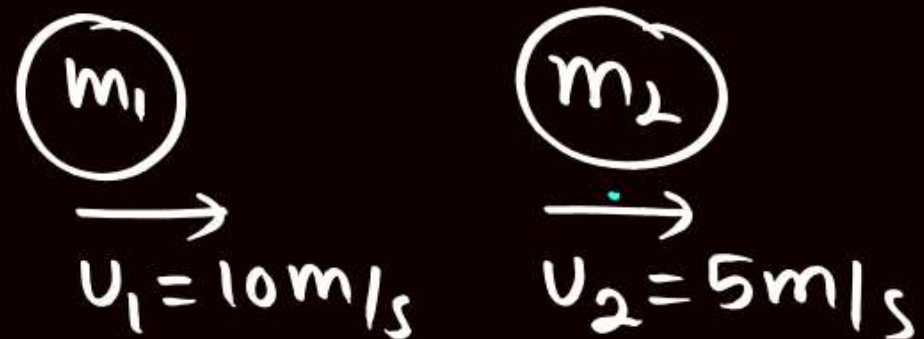
$$e = ?$$

Sol

$$e = \frac{\text{Velocity of S}}{\text{velocity of A}}$$

$$= \frac{V_2 - V_1}{u_1 - u_2} = \frac{10 - 10}{10 - 5} = \frac{0}{5} = 0$$

Q.2



Sol

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$= \frac{10 - 5}{10 - 5} = \frac{5}{5} = 1$$

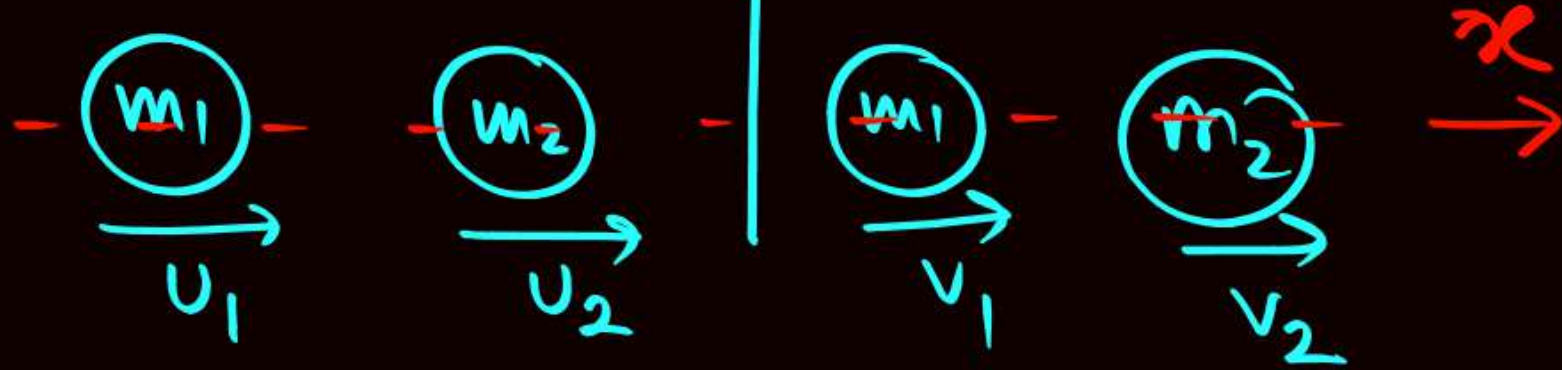
Perfectly elastic
collision

$e \rightarrow$ do not have unit

Note

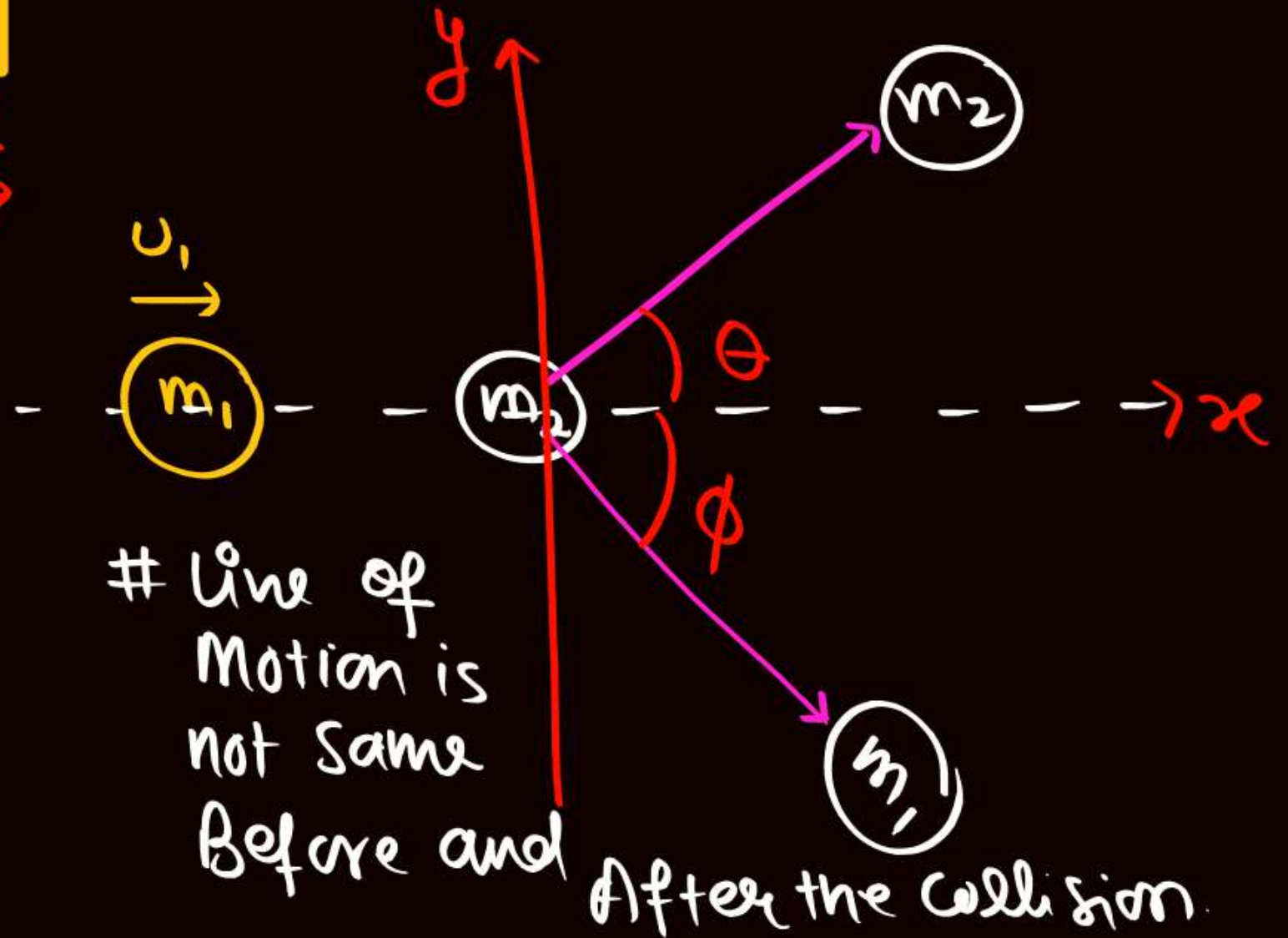
Collision

1D - collision (Head on collision)



Line of Motion is Same
Before and After the
collision

2D - collision

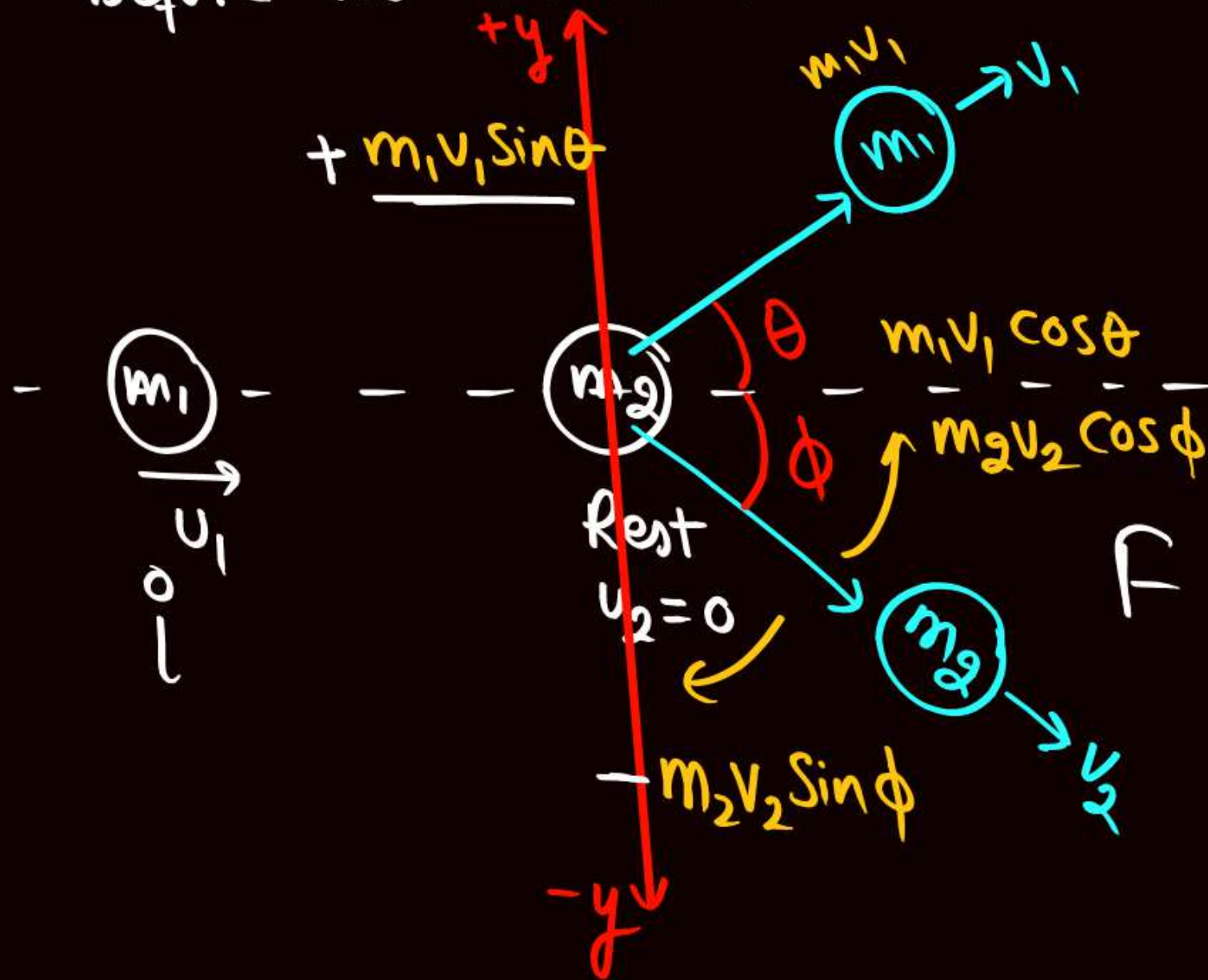


Line of Motion is
not Same
Before and After the collision.

Special case Neet

2D - Collision (Elastic collision)

When the line of motion is different After and before the collision



Elastic collision $\rightarrow P \rightarrow$ conserve
 $\rightarrow K.E \rightarrow$ conserve

① Momentum Conservation

$\rightarrow$ x-axis

$$m_1 u_1 + m_2 u_2^0 = m_1 v_1 \cos \theta + m_2 v_2 \cos \phi$$

$$m_1 u_1 = m_1 v_1 \cos \theta + m_2 v_2 \cos \phi \quad \text{--- (I)}$$

$\rightarrow$ y-axis

$$0 = m_1 v_1 \sin \theta - m_2 v_2 \sin \phi$$

$$m_2 v_2 \sin \phi = m_1 v_1 \sin \theta \quad \text{--- (II)}$$

② Kinetic ^{scalar} Energy Conservation

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^0 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad \text{--- (3)}$$

Power

$$1 \text{ km} = 1000 \text{ m}$$

Another famous formula of
Power

$$P = \frac{W}{t}$$

$$AB \cos \theta = \vec{A} \cdot \vec{B}$$

Power is Rate of
Doing work

$$\# P = \frac{W}{t} \xrightarrow{\text{J}} \frac{\text{J}}{\text{s}} = \text{watt}$$

• S.I unit Power = watt

• Another
unit power
↓

Horse power (Hp)

$$W = \vec{F} \cdot \vec{x}$$

Board
2023

$$\underline{1 \text{ Hp}} = 746 \text{ watt}$$

$$P = \frac{F \cdot x \cos \theta}{t}$$

$$P = FV \cos \theta$$

$$P = \vec{F} \cdot \vec{v}$$

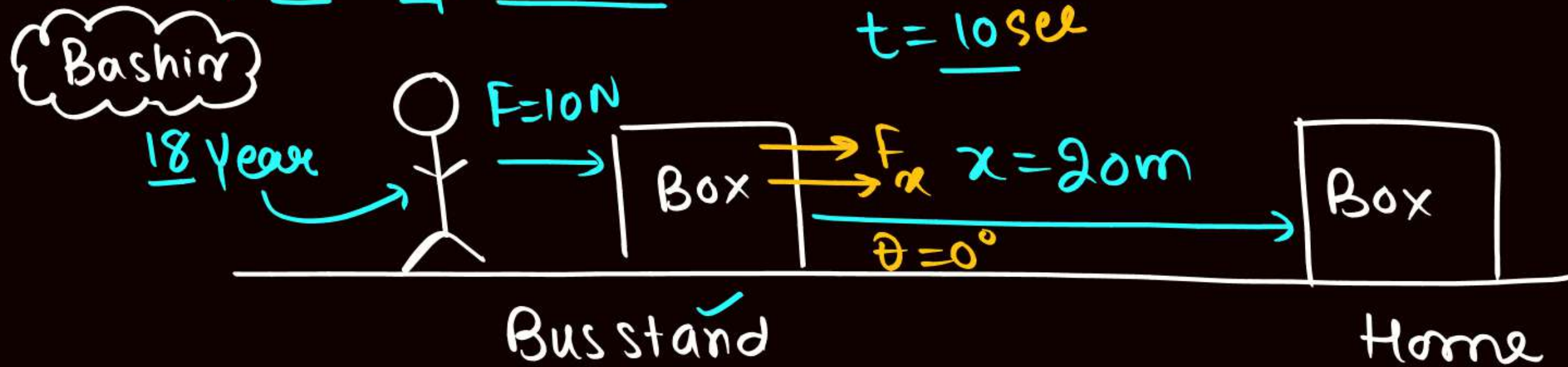
$$P = \frac{\text{Work}}{T} \quad \checkmark$$

$$P = \frac{\text{Energy}}{T}$$

$$P = \frac{K \cdot \epsilon}{t}$$

$$P = \frac{P \cdot \epsilon}{T}$$

Feel of Power



Sol

$$P = \frac{W}{T}$$

$$P = \frac{200 \text{ J}}{10 \text{ s}}$$

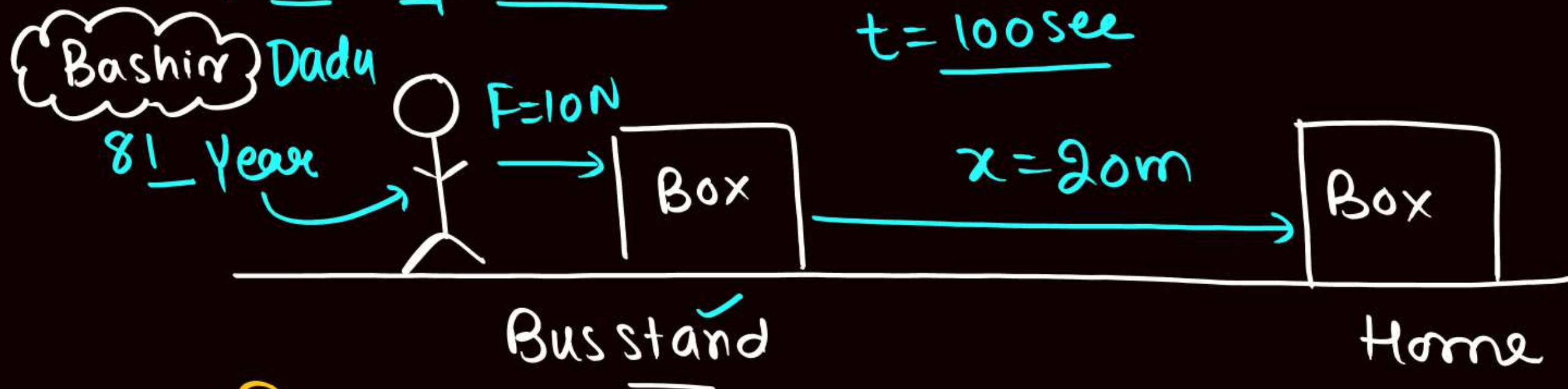
$$= 20 \text{ J/s} = 20 \text{ watt}$$

$$W = Fx \cos \theta$$
$$W = 10 \times 20 \cos 0^\circ$$

$$W = 10 \times 20 = 200 \text{ J}$$

Dadu

Feel of Power



$$P = \frac{W}{T}$$

$$\begin{aligned} W &= Fx \\ &= 10 \times 20 \\ &= 200\text{ J} \end{aligned}$$

$$\begin{aligned} \downarrow P &= \frac{200\text{ J}}{100\text{ s}} = 2\text{ J/s} \\ &= \underline{2\text{ watt}} \end{aligned}$$

Q Derive the Relationship b/w P and $K.E$ Board RS:-1001-

Neet
JEE
Sol
Sol

$$K.E = \frac{1}{2}mv^2$$

$$P = mv$$

momentum

JKBOSE
Kinetic
Energy

2016

2024 } MCQ
2025 }

$$K.E = \frac{1}{2}m\left(\frac{P}{m}\right)^2$$

$$\frac{P}{m} = v$$

$$K.E = \frac{P^2}{2m}$$

$$2m K.E = P^2 \text{ Tonzella}$$

$$\sqrt{2m K.E} = P$$

Tenzella

$$K.E = \frac{1}{2}m \frac{P^2}{m^2}$$

$$K.E = \frac{P^2}{2m}$$

Gabbar ✓

Numerical Series

PYQs

YE ASAAN HAI [SUMMER SERIES-A 2026]

The work done by a conservative force moving a particle around a closed path is:

- A** Always positive
- B** Always negative
- C** Zero
- D** Depends on the path

TUM SE NA HO PAYEGA? [SUMMER SERIES-A 2026]

An elevator weighing 600 kg is to be lifted up at a constant velocity of 0.5 m/s. What should be the minimum horse power of the motor to be used?

YE EASY HAI? [SUMMER SERIES-A 2026]

What is meant by a collision? Discuss two types of collision.

YE EASY HAI? [SUMMER SERIES-B 2026]

If a body's velocity changes direction but not magnitude, the motion is:

- A** Rectilinear
- B** Uniform linear
- C** Uniform circular
- D** Projectile

YE EASY HAI? [SUMMER SERIES-B 2026]

Which of the following forces is an example of conservative forces?

- A** Air resistance
- B** Viscous force
- C** Gravitational force
- D** None of these

TUM SE NA HO PAYEGA? [SUMMER SERIES-B 2026]

To lift a mass of 30 kg to a height of 15 cm, calculate the work done against gravity.

YE EASY HAI? [SUMMER SERIES-C 2026]

Which of the following is an example of a non-conservative force?

- A** Gravitational force
- B** Spring force
- C** Frictional force
- D** Electrostatic force

CAN YOU SOLVE THIS? [SUMMER SERIES-A 2025]

What is meant by restoring force and force constant?

DIFICULT HAI? QUESTION [SUMMER SERIES-A 2025]

A body of mass 10 kg initially at rest is subjected to a force of 40 N. What is the kinetic energy acquired by the body at the end of 10 sec?

YE EASY HAI? [SUMMER SERIES-A 2025]

Distinguish between elastic and inelastic collisions.

YE WALA ASAN HAI [SUMMER SERIES-C 2025]

The correct relationship between Calorie and Joule is

- A** 1 Calorie = 4.2 Joules
- B** 1 Calorie = $1/4.2$ Joules
- C** 0.24 Calorie = 1 Joule
- D** Both (A) and (C)

TENSAN HONE LAGA [JKBOSE JAMMU-2025-A]

A force of 50 N acts on a body and moves it to a distance of 4 m on a horizontal surface. Calculate work done if the direction of force is at an angle of 60° to the horizontal surface.

CAN YOU SOLVE THIS? [JKBOSE JAMMU-2025-A]

Give two points of difference between conservative and non-conservative forces.

YE WALA EASY HAI [JKBOSE JAMMU-2025-A]

What is Collision? Define two types of collision.

TUMSE NHI HOGA? [JKBOSE JAMMU-2025-B]

An elevator weighing 750 kg is to be lifted up at constant velocity of 0.3 m/s. What should be the minimum horse power of the motor to be used?

CAN YOU SOLVE THIS [JKBOSE JAMMU-2025-B]

Define power and energy. Give their units.

YE TO ASAN HAI [JKBOSE JAMMU-2025-B]

What is collision? Define two types of collision.

CAN YOU SOLVE THIS? [JKBOSE JAMMU 2025-C]

A force $F = (10 + 0.5x)$ acts on a particle in x direction, where F is in newton and x is in metre. Find the work done by this force during a displacement from $x = 0$ to $x = 2\text{m}$.

YE EASY HAI? [JKBOSE JAMMU-2025-C]

How do potential energy of a spring vary with displacement?

YE KR PAOGE? [JKBOSE JAMMU-2025-C]

What is Collision? Define two types of collision.

YE TO ARAM SE HO JAYEGA [JKBOSE JAMMU 2024-X]

A light and a heavy body have equal kinetic energy. Which has greater momentum?

- A** Heavy body
- B** Light body
- C** Both have equal momentum
- D** Information is incomplete



CAN YOU SOLVE THIS? [JKBOSE JAMMU 2024-X]

Define Work and Power. Give their S.I. units.

AAO SOLVE KRTE HAI [JKBOSE JAMMU 2024-X]

A force vec $\vec{F} = (2 + x)\text{N}$ acts on a particle in x -direction in metre. Find the work done by this force during a displacement from $x = 1\text{m}$ to $x = 2\text{m}$.

UE TO EASY HAI? [JKBOSE JAMMU 2024-Y]

A light and a heavy body have equal momentum. Which has greater kinetic energy?

- A** Heavy body
- B** Light body
- C** Both have equal momentum
- D** Information is incomplete



CAN YOU SOLVE THIS? [JKBOSE JAMMU 2024-Y]

Derive work-energy theorem.

YE MUSHKIL HAI KYA? [JKBOSE JAMMU 2024-Y]

A particle moves with a velocity of $\vec{v} = (6\hat{i} - 4\hat{j} + 3\hat{k})\text{m/s}$ under the influence of a constant force:

$$\vec{F} = (20\hat{i} + 15\hat{j} - 5\hat{k})\text{N}$$

Find the instantaneous power applied to the particle.

CAN YOU SOLVE THIS? [JKBOSE JAMMU 2024-Z]

Define coefficient of restitution. How does it define elastic and inelastic collision?

YE MUSHKIL HAI? [JKBOSE JAMMU 2024-Z]

A force $\vec{F} = (2 + x)\text{N}$ acts on a particle in x -direction in metre. Find the work done by this force during a displacement from $x = 1\text{m}$ to $x = 2\text{m}$.

**SUCCESS IS NOT PERMANENT AND
FAILURE IS NOT FINAL
NEVER STOP WORKING AFTER SUCCESS
AND, NEVER STOP TRYING AFTER
FAILURE**

HOMEWORK



H/w → GNS

Thank
You

